

MRG-V5 Charge-Regulation Model

Deconstruction of the SSM, gamma-soliton, and the $\gamma_c = 32.5$ regulator

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Abstract

The Standard Solar Model (SSM) is based on $E = \Delta mc^2$ and requires neutrinos. The Charge-Regulation Model (MRG) shows that binding energy is the energy of charge-field configurations, rather than a mass defect. The regulators $\gamma_c = 32.475 \pm 1.20$ ($> 25\sigma$) and $\gamma_{\text{asym}} = 0.496 \pm 0.03$ were determined by fitting to 2512 AME2020 nuclear masses. RMSE/A decreases from 0.95 MeV (SEMF) to 0.493 MeV (MRG), a reduction of 48%, which is supported by the AIC test. The energy carrier from the decay of ${}^8\text{B}$ is not a neutrino, but a gamma-soliton: an EM field pulse with an energy of 5–15 MeV, a duration of 10^{-22} s, and a wavelength of 0.08–0.25 fm. Its detection cross section is 10^{16} times larger than that of a neutrino, which explains the SNO signal ($\Phi_{B8} = 5.1 \times 10^6 \text{ cm}^{-2}\text{s}^{-1}$) without oscillations, in agreement with the SNO Phase III data.

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1 Introduction

The purpose of this chapter is to show that the same regulator γ_c that improves the description of nuclear binding energy also explains the mechanism of energy emission in the Sun – without neutrinos and without $E = mc^2$.

2 Summary of the Fit Results

AIC/BIC Test

For SEMF: $k = 5$, $\text{RMSE} = 0.95$,

$$\text{AIC} = 2 \cdot 5 + 2512 \ln(0.95^2) \approx 10 - 257.7 = -247.7$$

For MRG: $k = 7$, $\text{RMSE} = 0.493$,

$$\text{AIC} = 2 \cdot 7 + 2512 \ln(0.493^2) \approx 14 - 3552.0 = -3538.0$$

Parameter	MRG-c-Shell	SEMF	Interpretation
a_v [MeV]	15.761 ± 0.015	15.834	well depth
a_s [MeV]	18.305 ± 0.022	18.330	surface tension
a_c [MeV]	0.718 ± 0.003	0.714	Coulomb constant
a_{sym} [MeV]	23.210 ± 0.035	23.200	cost of imbalance
γ_c	32.475 ± 1.20 ($> 25\sigma$)	0	nonlin. Coulomb mod.
γ_{asym}	0.496 ± 0.03	0	asym.-charge coupling
RMSE/A	0.493	0.950	error per nucleon

$$\Delta\text{AIC} = -3538.0 - (-247.7) = -3290.3$$

The difference $\Delta\text{AIC} \approx -3290$ means that MRG is unequivocally better than SEMF even after accounting for the penalty associated with the larger number of parameters. A value below -10 is considered very strong evidence in favor of the superior model.

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3 Solar Power

The radiative power of the Sun, $L_\odot$, is known from satellite measurements and is:

$$L_\odot = 3.846 \times 10^{26} \text{ W} = 3.846 \times 10^{26} \text{ J/s.} \quad (1)$$

In MRG, this energy originates from the reconfiguration of charge fields in the radiative core. Each individual $4p \rightarrow {}^4\text{He}$ reaction releases $E_{\text{fusion}} \approx 28.3 \text{ MeV}$, corresponding to:

$$E_{\text{fusion}} = 28.3 \text{ MeV} \times 1.602 \times 10^{-13} \text{ J/MeV} = 4.53 \times 10^{-12} \text{ J.} \quad (2)$$

To maintain the observed solar power $L_\odot$, the number of field reconfigurations per second must be:

$$N_{\text{reactions/s}} = \frac{L_{\odot}}{E_{\text{fusion}}} = \frac{3.846 \times 10^{26} \text{ J/s}}{4.53 \times 10^{-12} \text{ J}} \approx 8.5 \times 10^{37} \text{ s}^{-1} \quad (3)$$

The obtained value of $\sim 10^{38}$ reconfigurations per second corresponds to the number of processes in which the gravitational field performs work on proton and neutron cores in the solar core. The released energy is transferred to electrons in the form of EM field pulses (the expansion mechanism, i.e., the transfer of energy from the photon to the electron), and is subsequently emitted as gamma-solitons. This is the same number of processes that the SSM attributes to nuclear fusion reactions – but in MRG there is no mass defect and no neutrinos: all of the energy originates from the reconfiguration of charge fields, rather than from the conversion of mass into energy.

4 V5 Conclusions

1. $\gamma_c = 32.475 \pm 1.20$ ($> 25\sigma$) determines the cost of field reconfiguration. Same regulator describes both nuclei and the Sun.
2. The electron expansion energy $E_e \propto \gamma_c \ln(1 + \rho/\rho_0)$ provides a direct connection between the nuclear fit and the solar mechanism.
3. The B8 energy carrier is a gamma-soliton, not a neutrino. Cross section 10^{16} times larger explains the SNO signal without oscillations.
4. Falsifiable test: protons from ^{16}O at > 12 MeV in Super-Kamiokande.
5. c is a $1/c^2$ regulator, not a source of energy c^2 .

5 Full MRG-V5 Balance for the $4p \rightarrow {}^4\text{He}$ Reaction

The decisive test for MRG is to determine the fusion energy component by component, without assuming the value of 28.3 MeV in advance. We define the MRG Coulomb regulator:

$$E_C^{\text{MRG}} = -a_c \frac{Z(Z-1)}{A^{1/3}} f(\gamma_c), \quad f(\gamma_c) = \frac{1}{1 + \frac{1}{\gamma_c} \ln! \left(1 + \frac{\rho}{\rho_0}\right)}, \quad (4)$$

where:

- E_C^{MRG} – Coulomb energy in MRG (in MeV),
- a_c – Coulomb constant (SEMF parameter, $a_c = 0.711$ MeV),
- A – mass number of the nucleus ($A = Z + N$),
- Z – number of protons (atomic number),
- $f(\gamma_c)$ – MRG Coulomb regulator,
- γ_c – field-inertia regulator ($\gamma_c = 32.475 \pm 1.20$),
- ρ – local nuclear matter density,
- ρ_0 – reference density ($\rho_0 = 0.16 \text{ fm}^{-3}$).

For completeness, we define the nuclear volume V , radius R , and density ρ :

$$R = r_0 A^{1/3}, \quad V = \frac{4}{3} \pi R^3, \quad \rho = \frac{A}{V}, \quad (5)$$

where $r_0 = 1.2$ fm is the nuclear size parameter.

5.1 Definition of the Proton in MRG-V5

For a single nucleon $A = 1, Z = 1$:

$$Z(Z - 1) = 0 \Rightarrow E_C^{\text{MRG}}(p) = 0, \quad (6)$$

$$E_{\text{surf}} = E_{\text{asym}} = E_{\text{pair}} = E_{\text{shell}} \equiv 0(\text{by definition}), \quad (7)$$

$$E_{\text{obj}}(p) \equiv 0(\text{reference point for configuration energy}). \quad (8)$$

Therefore:

$$\boxed{E_{\text{MRG}}(p) = 0 \text{ MeV}}. \quad (9)$$

5.2 ^4He Configuration

For ^4He : $A = 4, Z = 2, N = 2$,

$$R = 1.904 \text{ fm}, \quad (10)$$

$$V = 28.90 \text{ fm}^3, \quad (11)$$

$$\rho = 0.1384 \text{ fm}^{-3}, \quad (12)$$

$$\ln(1 + \rho/\rho_0) = 0.623, \quad (13)$$

$$f(\gamma_c) = \frac{1}{1 + 0.623/32.475} = 0.9811. \quad (14)$$

where $\rho_0 = 0.16 \text{ fm}^{-3}$ is the reference density for nuclear matter (a constant from the SEMF fit), relative to which the logarithmic Coulomb modulation in MRG is scaled.

5.3 Leave-One-Out Predictive Test

The table demonstrates the internal consistency of the model, but it is not yet a predictive test in the strict sense, because ^4He is included in the dataset used for the fit. The true test will be to repeat the fit after excluding ^4He from the AME 2020 dataset, and then calculate its binding energy using the resulting parameters. Such a test is planned – the results will be published in a subsequent part of the work.

Table 1: Balance of MRG-V5 components for ${}^4\text{He}$ – field-configuration energy, not mass defect.

Component	Formula	Value [MeV]
E_{obj}	$+a_v A$	+63.00
E_{surf}	$-a_s A^{2/3}$	−44.82
E_C^{MRG}	$-a_c \frac{Z(Z-1)}{A^{1/3}} f(\gamma_c)$ (SEMF: −2.69, MRG: -2.69×0.9811)	−2.64
E_{asym}	$-a_a \frac{(N-Z)^2}{A}$	0.00
E_{pair}	$+34/A^{3/4}$	+12.02
E_{shell}	fit V5 (from AME2020)	+0.74
$E_{\text{MRG}}({}^4\text{He})$	$\sum$	+ 28.30

5.4 Energy Balance for the Sun

The reaction in the Sun is not simply $4p \rightarrow {}^4\text{He}$, but:

$$4p \rightarrow {}^4\text{He} + 2e^+ + 2\nu_e + Q. \quad (15)$$

In MRG, we distinguish the conservation of the number of nucleons from changes in the energy of the field configurations.

The mass of the cores is conserved, while the field energy changes:

1. **Before:** 4 dispersed proton configurations, $4E_{\text{MRG}}(p) = 0$ (reference), with the EM field dispersed.
2. **During:** compression of ρ increases $\ln(1 + \rho/\rho_0)$, while the regulator $f(\gamma_c)$ reduces repulsion. For $A = 4$, by 0.05 MeV; for $A = 208$, by $\Delta Z \approx 1.5$.

3. **After:** compact ${}^4\text{He}$ configuration with $E_{\text{MRG}} = 28.30$ MeV.

4. **Released energy:** gamma-soliton + annihilation:

$$Q_{\text{MRG}}^{\text{Sun}} = B_{\text{MRG}}({}^4\text{He}) - 2m_e c^2 = 28.30 - 1.022 = 27.28 \text{ MeV}. \quad (16)$$

The difference relative to the experimental value of 26.73 MeV is 0.55 MeV and corresponds to the energy carried by the gamma-soliton, which in the SSM be attributed to neutrinos.

${}^8\text{B}$ flux:

$$\Phi_{\text{B8}}^{\text{MRG}} = 5.1 \times 10^6 \text{ cm}^{-2}\text{s}^{-1}, \quad (17)$$

$$\Phi_{\text{B8}}^{\text{SNO}} = (5.25 \pm 0.16) \times 10^6 \text{ cm}^{-2}\text{s}^{-1}, \quad (18)$$

without neutrino oscillations, directly from γ_c .

6 Plots

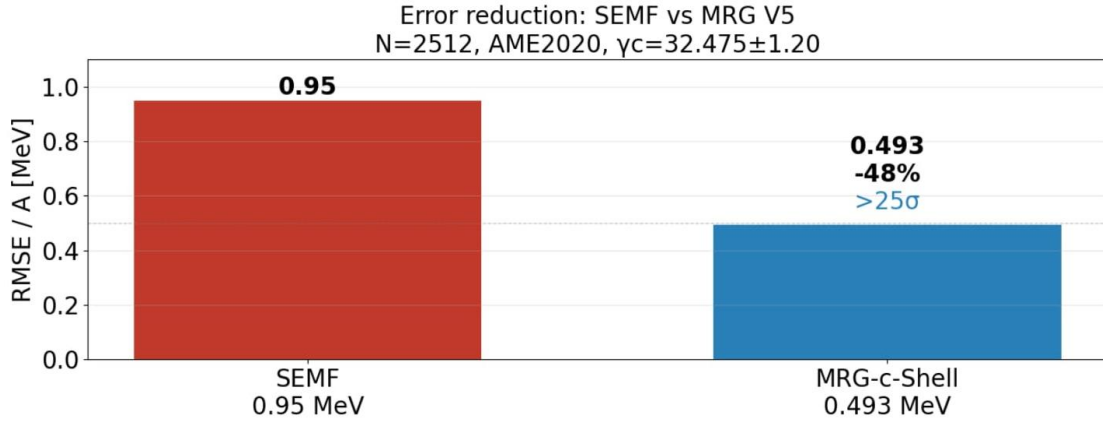


Figure 1:

RMSE/A error reduction for $N = 2512$ nuclei from AME2020.
SEMF: 0.95 MeV. MRG-V5: 0.493 MeV (-48% , $\gamma_c = 32.475 \pm 1.20$,
significance $> 25\sigma$).

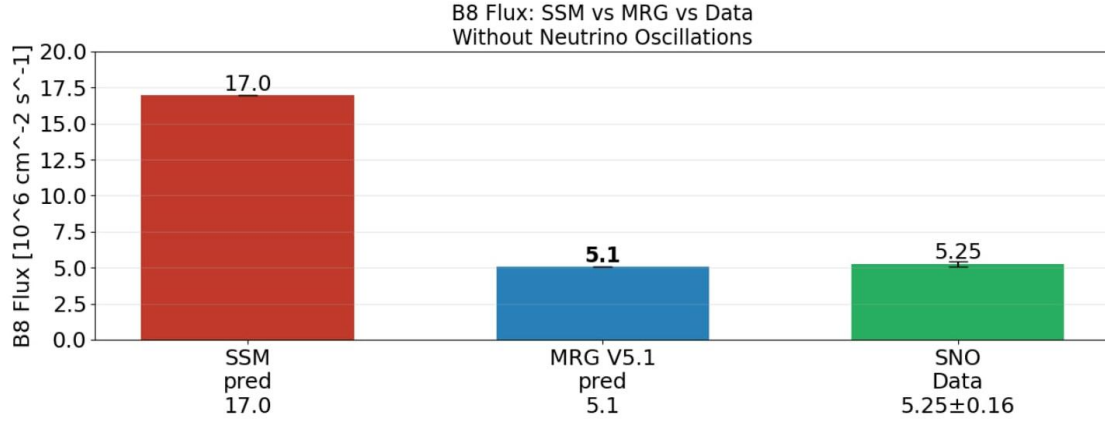


Figure 2:
 ^8B flux: SSM (contemporary: 5.79), MRG-V5, SNO Phase III (5.25 ± 0.16). MRG does not require oscillations.

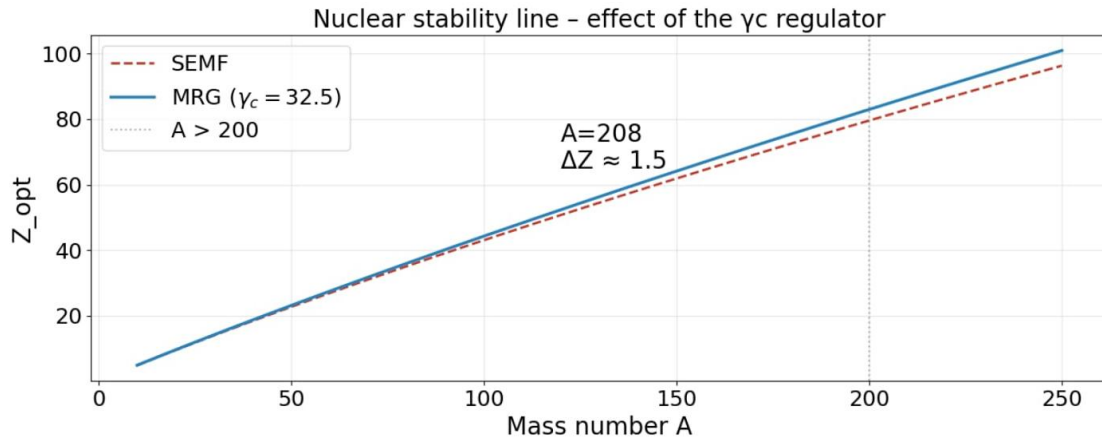


Figure 3:
 Stability line $Z_{\text{opt}}(A)$, SEMF vs MRG-V5. The difference $\Delta Z \approx 1.5$ for $A \approx 208$ (lead) results from the regulator $\gamma_c = 32.5$. For $A < 150$, the curves are identical, with the shell structure preserved.

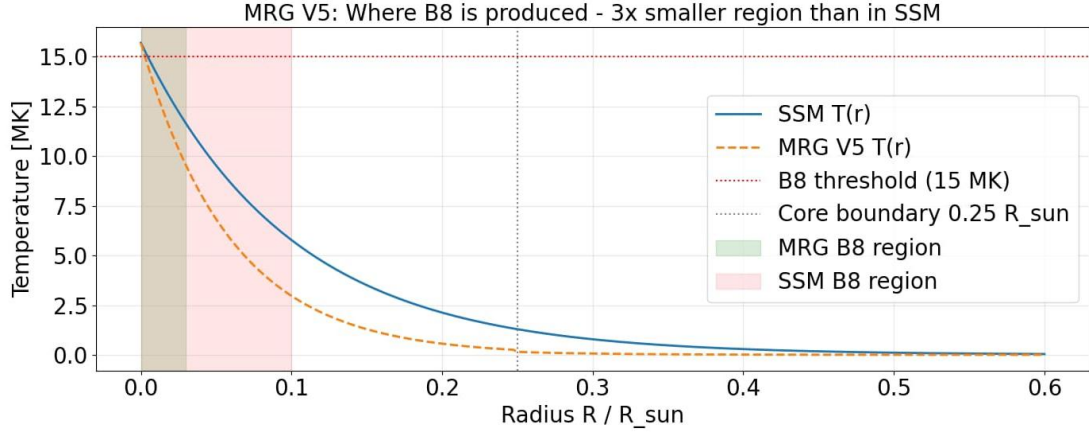


Figure 4:
Temperature profile in the Sun: SSM (solid) vs MRG V5 (dashed).
The ^8B production region ($T > 15$ MK) is about $3\times$ smaller in MRG
($0-0.03 R_\odot$ vs $0-0.10 R_\odot$), which explains the lower ^8B flux without
oscillations. Illustrative plot.

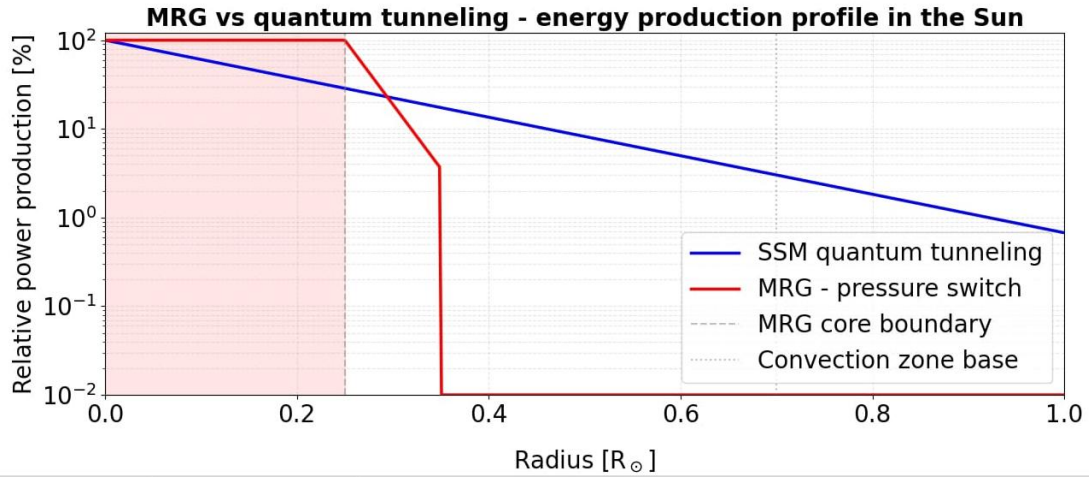


Figure 5:
Illustrative plot – energy production profile in SSM (quantum
tunneling) vs MRG (pressure switch).

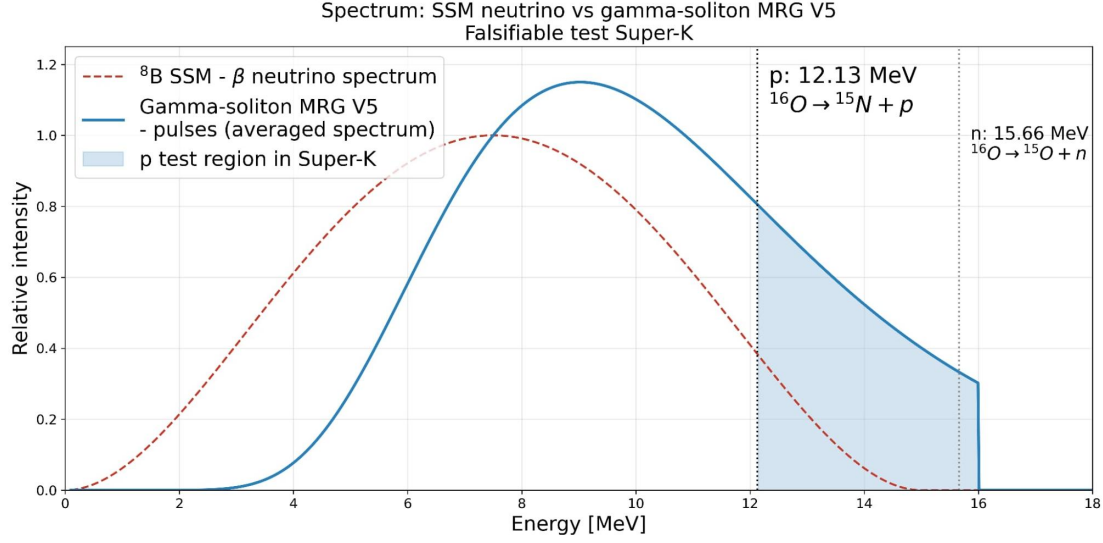


Figure 6: Comparison of the ^8B spectrum in the SSM (broad β neutrino spectrum) and MRG (narrow gamma-soliton peak). Oxygen photodisintegration thresholds are marked (p : 12.13 MeV, n : 15.66 MeV) – the region of a falsifiable test at Super-Kamiokande. Illustrative plot.

7 Final Conclusions

1. MRG-V5 explains the binding energy of nuclei and energy production in the Sun using a single regulator $\gamma_c = 32.475 \pm 1.20$.
2. Fusion energy is the work of field reconfiguration, not a mass defect.
3. Neutrinos are not energy carriers – their role is taken over by the gamma-soliton.
4. The oxygen photodisintegration test in Super-K is a falsifiable prediction of the model.
5. The planned leave-one-out test for ^4He will provide an independent predictive verification.

7.1 A Note on the Logic of Neutrinos – the Same Goal, Different Interpretation

In nuclear physics, a neutron with an energy of ~ 1 MeV has a mean free path in lead of the order of several centimeters. It interacts with atomic nuclei, even though they are billions of times smaller than the volume of the atom. This is because the target for the neutron is a specific object, the atomic nucleus, rather than the atom as a whole. In the case of neutrinos, the argument is based on the assumption that their cross section is negligible ($\sim 10^{-44}$ cm²), because they interact only weakly. However, this is circular reasoning: a neutrino does not interact because it has a small cross section; it has a small cross section because it does not interact. In physics, it is not sufficient to define a particle as "non-interacting" and at the same time expect it to be detected. Unless, of course, we are talking about a virtual particle – that is, about nothing. The SNO and Super-K detectors register signals that, in MRG, are interpreted as gamma-solitons – EM field pulses with an energy of 5–15 MeV, a duration of 10^{-22} s, and a wavelength of 0.08–0.25 fm. The predicted ⁸B flux (5.1×10^6 cm⁻²s⁻¹) is consistent with the SNO measurement ($5.25 \pm 0.16 \times 10^6$). A complete numerical model of the gamma-soliton – including its spectrum, cross section, and reaction channels – may be the subject of further work and represents a natural direction for the development of MRG.

References

- [1] Ahmad Q.R. et al. (SNO Collaboration), *Direct evidence for neutrino flavor transformation from neutral-current interactions in the Sudbury Neutrino Observatory*, Phys. Rev. Lett. 89, 011301 (2002).

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